

PROJECT FINAL REPORT

USED CAR DEPRECIATION RATE

Machine Learning & Data Science for Economics

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PROJECT OVERVIEW & SETUP

OBJECTIVE

We analyze the depreciation rate of used vehicles to benefit US buyers. By grouping vehicles by make and body type, we calculate specific depreciation curves for accurate market valuation.

DATASET STATS

Total Valid Data: 936,723 records.

Features: 23 meaningful attributes.

Robustness: Ensures 40+ records per group/year for a less biased normal distribution of prices.

| DATA CLEANING PIPELINE

FILTER

Removed 99% missing features (e.g., 'bed') and redundant IDs.



CONSOLIDATE

Merged duplicate VINs and imputed missing specs within cohorts.



STANDARDIZE

Grouped by Engine, Make, Model to fix minor VIN variations.



SPLIT

Time-based Train/Test split.
Cutoff date: **2020-09-17**.

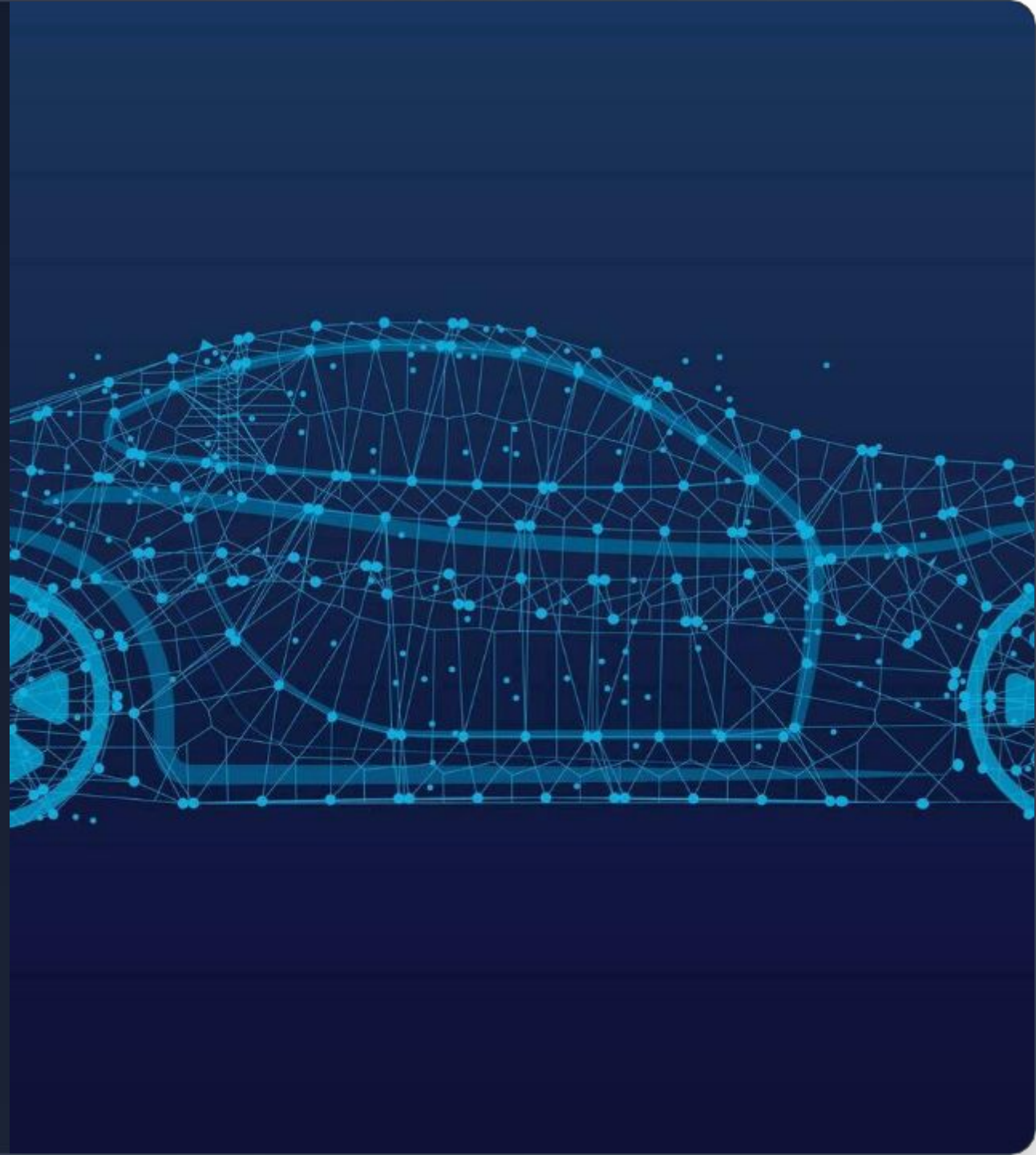


| MODELING APPROACH

LIGHTGBM FRAMEWORK

We selected Light Gradient Boosting Machine over traditional regression or large ensemble forests.

- ✓ **Efficiency:** Substantially faster training with lower memory usage.
- ✓ **Scalability:** Handles 870k+ listings efficiently.
- ✓ **Complexity:** Captures non-linear relationships in high-dimensional categorical data (brand, engine size, luxury status).



| TRAINING OPTIMIZATION

1,642

OPTIMAL ITERATIONS

Two Key Hyperparameters

num_leaves: lets the trees learn more detailed brand- and trim-specific depreciation curves but risks overfitting.

learning_rate: improves generalization but requires more boosting rounds.

EARLY STOPPING STRATEGY

Initial training capped at 5,000 rounds. The model auto-halted at ~1,900, identifying **1,642** as the peak performance point.

We subsequently retrained the full dataset using this fixed iteration count to maximize learning without overfitting (validated against models with 2,500 rounds).

MODEL PERFORMANCE

874,441 final vehicles



**MEAN ABSOLUTE
ERROR(MAE)**

\$1,686.3



\$1,668.7

average dollar distance
from the true price



R-SQUARED (R^2)

0.970 → 0.971

Explains 97.1% of price variance

rMAPE

6.8% → 6.5%

Off by 6.5% on
average(25.8k)



RMSE

\$2,528.4



\$2,506.5

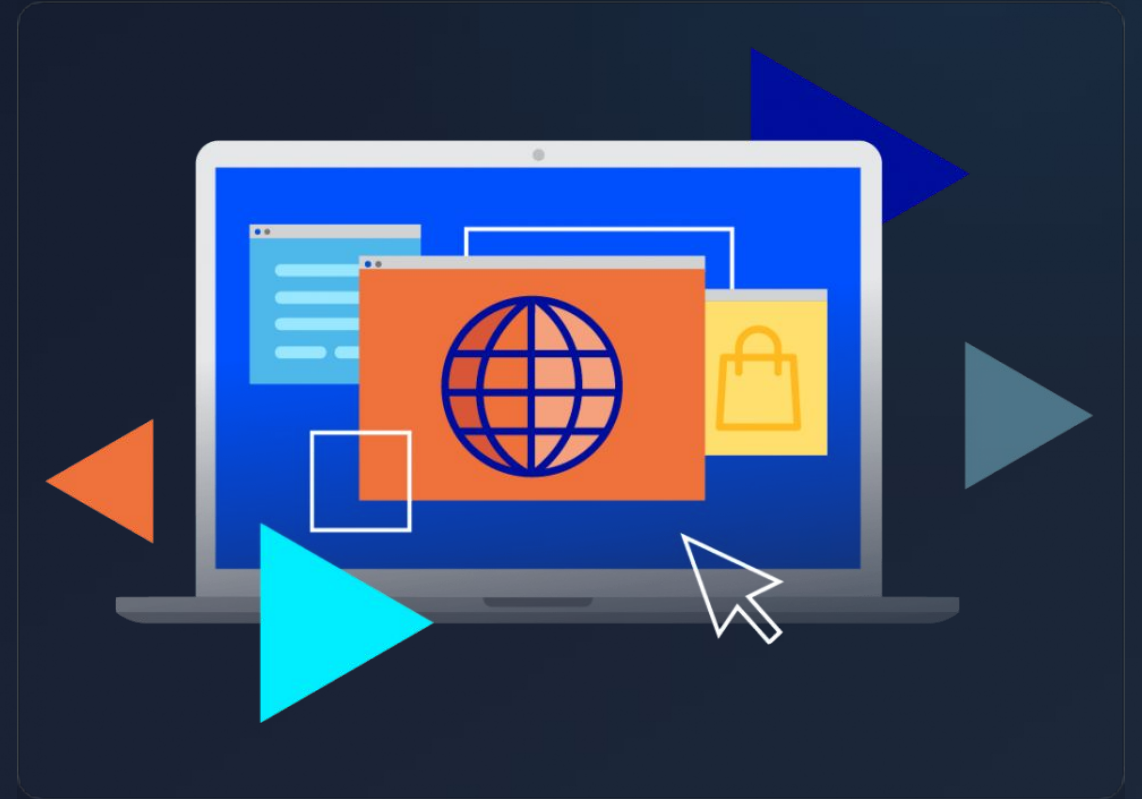
Penalize bigger mistakes
Modest gap to MAE indicates few
severe outliers

| INTERACTIVE WEB APPLICATION

STREAMLIT INTERFACE

An intuitive tool for users to explore vehicle valuation.

- 🔹 **Smart Cascading Filters:** Selecting a "Make" automatically updates available "Models" to prevent invalid inputs.
- 🔧 **Handling Unknowns:** Inputs marked "Not in List" or blank default to the median value of known features.



| LIMITATIONS & FUTURE WORK

DISTRIBUTIONAL SKEW

Vehicles older than 30 years and ultra-luxury models create disproportionate variance.

Solution: Implement segmented models or capped loss functions for extreme-value segments.

DATA SPARSITY

Niche trims and low-volume manufacturers currently produce larger residual errors.

Solution: Expand historical records and evaluate dedicated brand-specific models.

Q & A

Thank you for your attention.

Reference

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G. Ke et al., "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," Advances in Neural Information Processing Systems, pp. 3149–3157, 2017.



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